

The role of latitude, ultraviolet radiation exposure and vitamin D in childhood asthma and hayfever: an Australian multicenter study.

Observations of increasing allergy prevalence with decreasing distance from the Equator and positive associations with ambient ultraviolet radiation have contributed to a growing interest in the possible role of vitamin D in the etiology of allergy. The aims of this study were to describe any latitudinal variation in the prevalence of childhood allergy in Australia and to evaluate, in parallel, the individual associations between ultraviolet radiation (UVR)- and vitamin D-related measures and hayfever asthma and both conditions. Participants were population-based controls who took part in a multicenter case-control study, aged 18-61 yr and resident in one of four study regions ranging in latitude from 27°S to 43°S. Data were derived from a self-administered questionnaire, interview and examination by a research officer and biologic sampling. Latitude and longitude coordinates were geocoded from participants' residential locations and climatic data were linked to postcodes of current residence. Stored serum was analyzed for 25-hydroxyvitamin D concentrations and silicone rubber casts of the skin were used as an objective measure of cumulative actinic damage. There was an inverse latitude gradient for asthma (a 9% decrease per increasing degree of latitude); however, this pattern did not persist after adjusting for average daily temperature. There was no association between any of the UVR- or vitamin D-related measures and childhood asthma, but greater time in the sun in winter between the ages 6-15 yr was associated with an increase in the odds of having hayfever [adjusted odds ratios (OR) 1.29; 95% CI 1.01-1.63]. Oral supplementation with cod liver oil in childhood increased the odds of a history of having both asthma and hayfever (2.87; 1.00-8.32). Further investigation of the possible role of early vitamin D supplementation in the development of allergy is warranted. Our results also suggest that solar exposure during childhood may be important in allergic sensitization. Plausible explanations, including biologic mechanisms, exist for both observations. © 2010 John Wiley & Sons A/S.